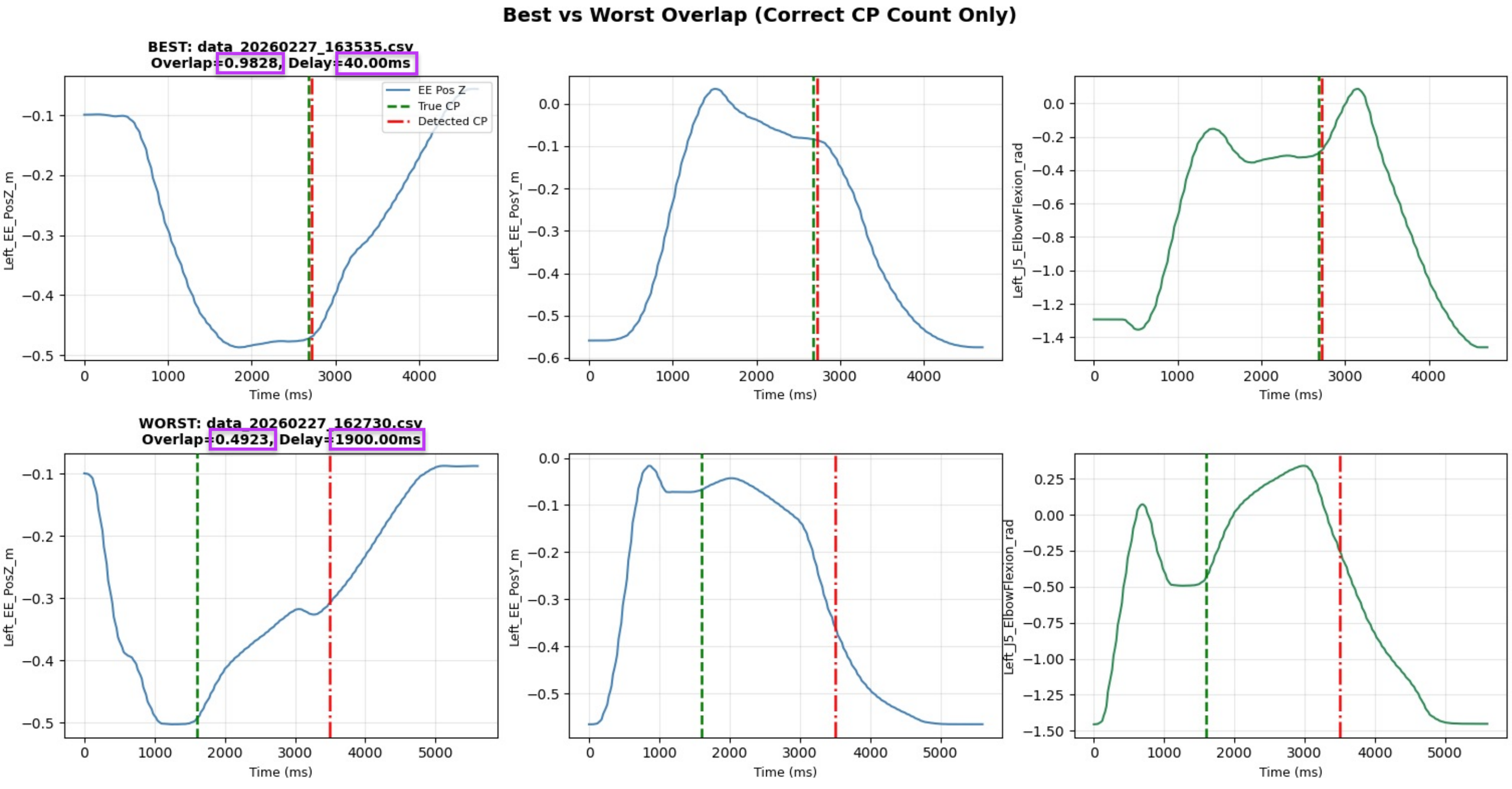
PELT Change Point Detection

RESULTS
27th February 2026


Best vs Worst Overlap (Correct CP Count Only)
BEST: data_20260227_163535.csv
Overlap=0.9828, Delay=40.00ms
EE Pos Z
True CP
Detected CP
Left_EE_PosZ_m
Left_EE_PosY_m
Left_J5_ElbowFlexion_rad
Time (ms)
WORST: data_20260227_162730.csv
Overlap=0.4923, Delay=1900.00ms

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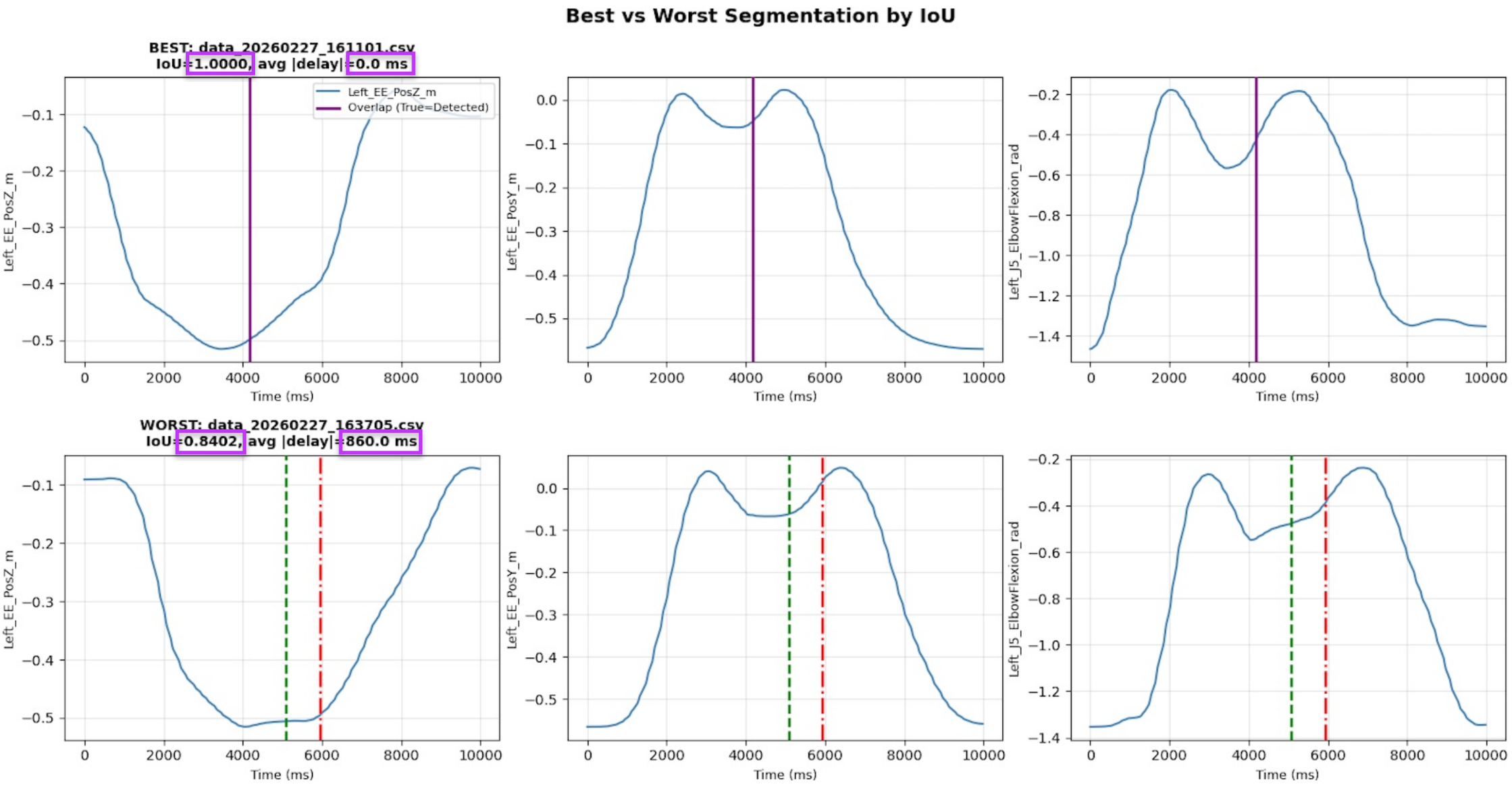
--- SUMMARY (Only samples with correct CP count) ---
Samples with correct CP count: 43/87 (49.4%)
Average Overlap: 0.7380
Average Delay (ms): 827.44

==================================================================
BEST overlap: data_20260227_163535.csv (overlap=0.9828)
WORST overlap: data_20260227_162730.csv (overlap=0.4923)
==================================================================
```

CNN Change Point Detection

RESULTS
2nd March 2026

- **Supervised CPD implementation**
    - **Model**: 1D CNN (3 Conv layers + BatchNorm/ReLU) with a 1x1 conv head.
    - **Optimizer**: Adam (lr=0.001)
    - **Training**:
        - K-fold cross validation (default 5 folds)
        - 50 Epochs
        - Batch size 4
        - Early-stopping patience 15
    - **Preprocessing**:
        - Interpolate sequences to fixed length (default 500)
        - Normalization: StandardScaler
        - Augmentation: temporal jitter, scaling, shift, warping, frame skipping
    - **CPD modes:**
        - Top K change points (default max_change_points=1)
        - OR Threshold-based on prediction probabilities (default threshold 0.5)
    - **Detection tolerance**: Window of samples to detect True Positives (default 20).
    - **Detection Metrics**: precision, recall, F1 score
    - **Segmentation metric**: Per-state and overall IoU
    - **Timing**: Average absolute delay in CPD (in ms)
    - **Checkpoints**: Per-fold weights and best-fold model saved.
    - **Visualization Plots**:
        - Training-loss curves
        - Best/worst segmentation plots (2×3 figure)
        - Precision vs detection window plots.


Best vs Worst Segmentation by IoU
BEST: data_20260227_161101.csv
IoU=1.0000, avg |delay|=0.0 ms
Left_EE_PosZ_m
Overlap (True=Detected)
Left_EE_PosY_m
Left_J5_ElbowFlexion_rad
Time (ms)
WORST: data_20260227_163705.csv
IoU=0.8402, avg |delay|=860.0 ms

```
================================================================
Test Set Metrics - Overall Summary
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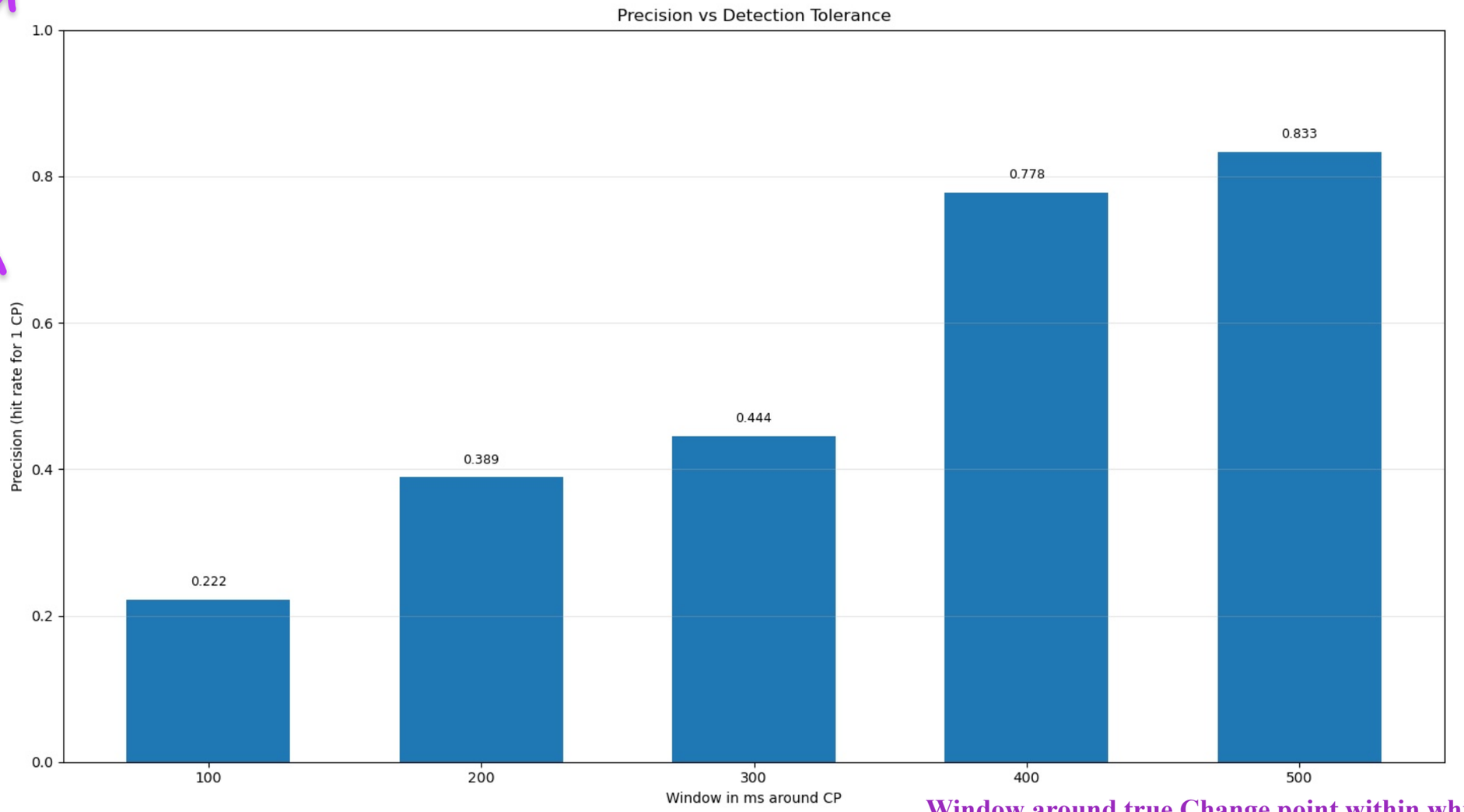
Metric                    Value
----------------------------------------------------
Average IoU               0.9400 ± 0.0000
IoU (GOING)               0.9398 ± 0.0000
IoU (RETURNING)           0.9403 ± 0.0000
----------------------------------------------------
Precision                 0.7778 ± 0.0000
Recall                    0.7778 ± 0.0000
F1 Score                  0.7778 ± 0.0000
----------------------------------------------------
```

Average overlap between predcted and actual labels for GOING and RETURNING

For a single change point, in this case, precision, Recall and F1 score are the same as the hit rate: proportion of data samples where the predicted CP is within given window

For a single change point, in this case, precision is the same as the hit rate: proportion of data samples where the predicted CP is within given window



Window around true Change point within which a detected change point is treated as a True Positive